

Challenge: Kangaroo Rats, Diets, and Body Size

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Set-up

1. Import the Data

```
[1] "1"
```

```
# A tibble: 35,549 x 9
```

	record_id	month	day	year	plot_id	species_id	sex	hindfoot_length	weight
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>
1	1	7	16	1977	2	NL	M	32	NA
2	2	7	16	1977	3	NL	M	33	NA
3	3	7	16	1977	2	DM	F	37	NA
4	4	7	16	1977	7	DM	M	36	NA
5	5	7	16	1977	3	DM	M	35	NA
6	6	7	16	1977	1	PF	M	14	NA
7	7	7	16	1977	2	PE	F	NA	NA
8	8	7	16	1977	1	DM	M	37	NA
9	9	7	16	1977	1	DM	F	34	NA
10	10	7	16	1977	6	PF	F	20	NA

```
# i 35,539 more rows
```

```
# A tibble: 24 x 2
```

	plot_id	plot_type
	<dbl>	<chr>
1	1	Spectab enclosure
2	2	Control
3	3	Long-term Krat Enclosure
4	4	Control

```

5      5 Rodent Exclosure
6      6 Short-term Krat Exclosure
7      7 Rodent Exclosure
8      8 Control
9      9 Spectab exclosure
10     10 Rodent Exclosure
# i 14 more rows

```

```
# A tibble: 54 x 4
```

	species_id	genus	species	taxa
	<chr>	<chr>	<chr>	<chr>
1	AB	Amphispiza	bilineata	Bird
2	AH	Ammospermophilus	harrisi	Rodent
3	AS	Ammodramus	savannarum	Bird
4	BA	Baiomys	taylori	Rodent
5	CB	Campylorhynchus	brunneicapillus	Bird
6	CM	Calamospiza	melanocorys	Bird
7	CQ	Callipepla	squamata	Bird
8	CS	Crotalus	scutalatus	Reptile
9	CT	Cnemidophorus	tigris	Reptile
10	CU	Cnemidophorus	uniparens	Reptile

```

# i 44 more rows

```

```
# A tibble: 5,400 x 26
```

	MSW3_ID	Scientific	MSWFamilyLatin	`Diet-Inv`	`Diet-Vend`	`Diet-Vect`
	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	1	Tachyglossus acule~	Tachyglossidae	100	0	0
2	2	Zaglossus attenbor~	Tachyglossidae	100	0	0
3	3	Zaglossus bartoni	Tachyglossidae	100	0	0
4	4	Zaglossus bruijni	Tachyglossidae	100	0	0
5	5	Ornithorhynchus an~	Ornithorhynch~	80	0	0
6	6	Caluromys philander	Didelphidae	20	0	0
7	7	Caluromys derbianus	Didelphidae	20	0	0
8	8	Caluromys lanatus	Didelphidae	20	0	0
9	9	Caluromysiops irru~	Didelphidae	20	0	0
10	10	Glironia venusta	Didelphidae	40	0	0

```
# i 5,390 more rows
```

```

# i 20 more variables: `Diet-Vfish` <dbl>, `Diet-Vunk` <dbl>,
# `Diet-Scav` <dbl>, `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>,
# `Diet-Seed` <dbl>, `Diet-Plant0` <dbl>, `Diet-Source` <chr>,
# `Diet-Certainty` <chr>, `ForStrat-Value` <chr>, `ForStrat-Certainty` <chr>,

```

```
# `ForStrat-Comment` <chr>, `Activity-Nocturnal` <dbl>,
# `Activity-Crepuscular` <dbl>, `Activity-Diurnal` <dbl>, ...
```

2. Tidy the Trait Table

```
[1] "2"
```

```
# A tibble: 5,400 x 12
```

```
  sci_name    mass_g `Diet-Inv` `Diet-Vend` `Diet-Vect` `Diet-Vfish` `Diet-
Vunk`
  <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 Tachyglos~ 3025      100      0      0      0      0
2 Zaglossus~ 8532.    100      0      0      0      0
3 Zaglossus~ 7180      100      0      0      0      0
4 Zaglossus~ 10140.    100      0      0      0      0
5 Ornithorh~ 1484.      80      0      0     20      0
6 Caluromys~ 229.      20      0      0      0     10
7 Caluromys~ 297      20      0      0      0     10
8 Caluromys~ 325      20      0      0      0     10
9 Caluromys~ 250      20      0      0      0      0
10 Glironia ~ 114      40      0      0      0     20
# i 5,390 more rows
# i 5 more variables: `Diet-Scav` <dbl>, `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>,
# `Diet-Seed` <dbl>, `Diet-Plant0` <dbl>
```

3. Match Portal Species to Traits

```
[1] "3a"
```

```
# A tibble: 31 x 5
```

```
  species_id sci_name      genus      species      taxa
  <chr>      <chr>      <chr>      <chr>      <chr>
1 AH      Ammospermophilus harrisi Ammospermophilus harrisi Rodent
2 BA      Baiomys taylori      Baiomys      taylori Rodent
3 DM      Dipodomys merriami      Dipodomys      merriami Rodent
4 DO      Dipodomys ordii      Dipodomys      ordii Rodent
5 DS      Dipodomys spectabilis      Dipodomys      spectabilis Rodent
6 DX      Dipodomys sp.      Dipodomys      sp. Rodent
7 NL      Neotoma albigula      Neotoma      albigula Rodent
8 NX      Neotoma sp.      Neotoma      sp. Rodent
9 OL      Onychomys leucogaster      Onychomys      leucogaster Rodent
```

```
10 OT      Onychomys torridus      Onychomys      torridus      Rodent
# i 21 more rows
```

```
[1] "3b"
```

```
# A tibble: 9 x 5
```

	species_id	sci_name	genus	species	mass_g
	<chr>	<chr>	<chr>	<chr>	<dbl>
1	AH	Ammospermophilus harrisi	Ammospermophilus	harrisi	NA
2	DX	Dipodomys sp.	Dipodomys	sp.	NA
3	NX	Neotoma sp.	Neotoma	sp.	NA
4	OX	Onychomys sp.	Onychomys	sp.	NA
5	PH	Perognathus hispidus	Perognathus	hispidus	NA
6	PX	Chaetodipus sp.	Chaetodipus	sp.	NA
7	RX	Reithrodontomys sp.	Reithrodontomys	sp.	NA
8	SX	Sigmodon sp.	Sigmodon	sp.	NA
9	UR	Rodent sp.	Rodent	sp.	NA

```
[1] "3c"
```

```
# A tibble: 24 x 16
```

	species_id	sci_name	genus	species	taxa	mass_g	`Diet-Inv`	`Diet-Vend`
	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	AH	Ammospermophilu~	Ammo~	harris~	Rode~	122	20	0
2	BA	Baiomys taylori	Baio~	taylori	Rode~	8	10	0
3	DM	Dipodomys merri~	Dipo~	merria~	Rode~	42	20	0
4	DO	Dipodomys ordii	Dipo~	ordii	Rode~	60.4	20	0
5	DS	Dipodomys spect~	Dipo~	specta~	Rode~	136.	20	0
6	NL	Neotoma albigula	Neot~	albigu~	Rode~	206.	10	0
7	OL	Onychomys leuco~	Onyc~	leucog~	Rode~	27.9	90	0
8	OT	Onychomys torri~	Onyc~	torrid~	Rode~	25.0	90	0
9	PB	Chaetodipus bai~	Chae~	baileyi	Rode~	26.4	10	0
10	PE	Peromyscus erem~	Pero~	eremic~	Rode~	23.6	50	0

```
# i 14 more rows
```

```
# i 8 more variables: `Diet-Vect` <dbl>, `Diet-Vfish` <dbl>, `Diet-Vunk` <dbl>,
#   `Diet-Scav` <dbl>, `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>,
#   `Diet-Seed` <dbl>, `Diet-Plant0` <dbl>
```

4. Build the Analysis Table

```
[1] "4"
```

```
# A tibble: 20,451 x 25
```

	record_id	month	day	year	plot_id	species_id	sex	hindfoot_length	weight
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>
1	1	7	16	1977	2	NL	M	32	NA
2	2	7	16	1977	3	NL	M	33	NA
3	3	7	16	1977	2	DM	F	37	NA
4	5	7	16	1977	3	DM	M	35	NA
5	7	7	16	1977	2	PE	F	NA	NA
6	13	7	16	1977	3	DM	M	35	NA
7	14	7	16	1977	8	DM	<NA>	NA	NA
8	16	7	16	1977	4	DM	F	36	NA
9	17	7	16	1977	3	DS	F	48	NA
10	18	7	16	1977	2	PP	M	22	NA

```
# i 20,441 more rows
```

```
# i 16 more variables: sci_name <chr>, genus <chr>, species <chr>, taxa <chr>,  
# mass_g <dbl>, `Diet-Inv` <dbl>, `Diet-Vend` <dbl>, `Diet-Vect` <dbl>,  
# `Diet-Vfish` <dbl>, `Diet-Vunk` <dbl>, `Diet-Scav` <dbl>,  
# `Diet-Fruit` <dbl>, `Diet-Nect` <dbl>, `Diet-Seed` <dbl>,  
# `Diet-Plant0` <dbl>, plot_type <chr>
```

5. Relative Abundance of Kangaroo Rats

```
[1] "5a"
```

```
# A tibble: 46 x 5
```

	plot_type	genus	species	n	rel_abund
	<chr>	<chr>	<chr>	<int>	<dbl>
1	Control	Dipodomys	merriami	6760	0.438
2	Control	Dipodomys	ordii	1671	0.108
3	Control	Dipodomys	spectabilis	1389	0.0900
4	Control	Chaetodipus	penicillatus	1108	0.0718
5	Control	Onychomys	torridus	1004	0.0651
6	Control	Chaetodipus	baileyi	789	0.0511
7	Control	Neotoma	albigula	610	0.0395
8	Control	Onychomys	leucogaster	471	0.0305
9	Control	Perognathus	flavus	424	0.0275
10	Control	Reithrodontomys	megalotis	403	0.0261

```
# i 36 more rows
```

```
[1] "5b"
```

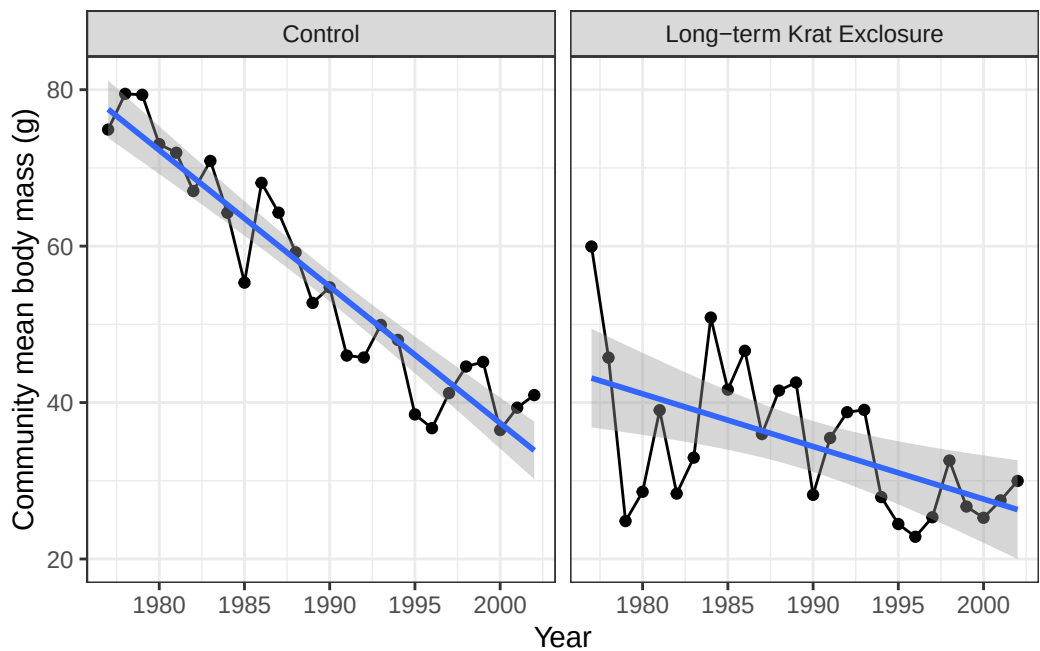
```
# A tibble: 2 x 2
  plot_type      prop_krats
  <chr>          <dbl>
1 Control      0.637
2 Long-term Krat Exclosure 0.0709
```

6. Body Size Through Time

```
[1] "6a"
```

```
# A tibble: 52 x 4
  year plot_type      n_captures mean_mass_g
  <dbl> <chr>          <int>      <dbl>
1  1977 Control      221        74.9
2  1977 Long-term Krat Exclosure    61        60.0
3  1978 Control      500        79.5
4  1978 Long-term Krat Exclosure    78        45.7
5  1979 Control      321        79.3
6  1979 Long-term Krat Exclosure    62        24.8
7  1980 Control      612        73.0
8  1980 Long-term Krat Exclosure   151        28.6
9  1981 Control      644        71.9
10 1981 Long-term Krat Exclosure   103        39.0
# i 42 more rows
```

```
[1] "6b"
```



7. Community Diet

[1] "7a"

A tibble: 204,510 x 5

	record_id	sci_name	plot_type	diet	diet_percent
	<dbl>	<chr>	<chr>	<chr>	<dbl>
1	1	Neotoma albigula	Control	Inv	10
2	1	Neotoma albigula	Control	Vend	0
3	1	Neotoma albigula	Control	Vect	0
4	1	Neotoma albigula	Control	Vfish	0
5	1	Neotoma albigula	Control	Vunk	0
6	1	Neotoma albigula	Control	Scav	0
7	1	Neotoma albigula	Control	Fruit	0
8	1	Neotoma albigula	Control	Nect	0
9	1	Neotoma albigula	Control	Seed	20
10	1	Neotoma albigula	Control	Plant0	70

i 204,500 more rows

[1] "7b"

```
# A tibble: 20 x 3
```

	plot_type	diet	mean_percent
	<chr>	<chr>	<dbl>
1	Control	Fruit	13.5
2	Control	Inv	25.3
3	Control	Nect	0
4	Control	Plant0	26.4
5	Control	Scav	0.382
6	Control	Seed	34.4
7	Control	Vect	0
8	Control	Vend	0
9	Control	Vfish	0
10	Control	Vunk	0
11	Long-term Krat Exclosure	Fruit	3.72
12	Long-term Krat Exclosure	Inv	21.8
13	Long-term Krat Exclosure	Nect	0
14	Long-term Krat Exclosure	Plant0	20.4
15	Long-term Krat Exclosure	Scav	1.15
16	Long-term Krat Exclosure	Seed	52.9
17	Long-term Krat Exclosure	Vect	0
18	Long-term Krat Exclosure	Vend	0
19	Long-term Krat Exclosure	Vfish	0
20	Long-term Krat Exclosure	Vunk	0

```
[1] "7c"
```

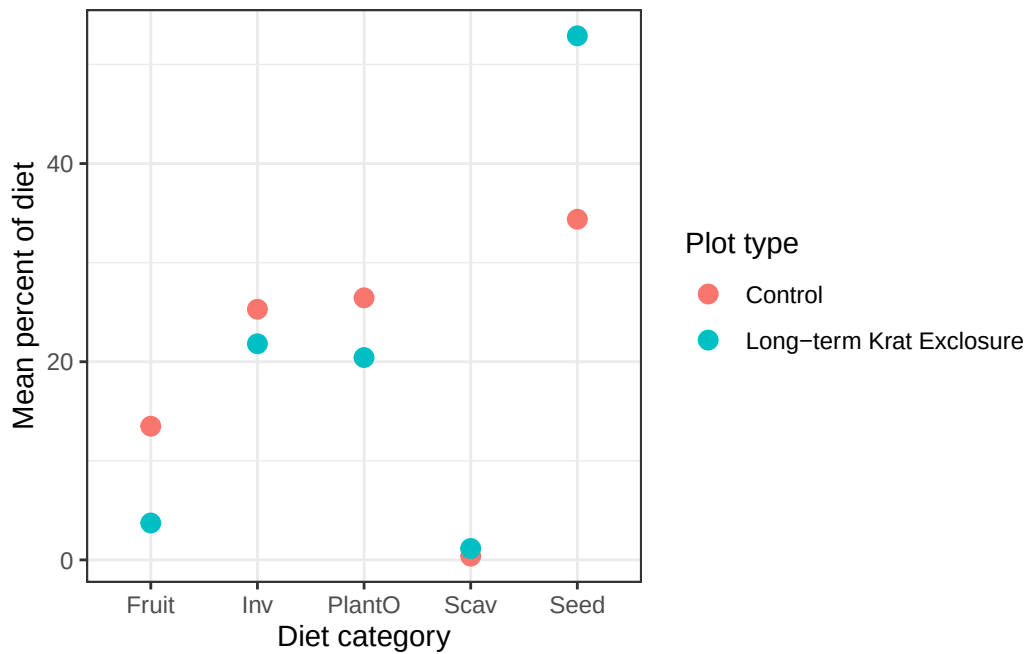
```
# A tibble: 10 x 3
```

	plot_type	diet	mean_percent
	<chr>	<chr>	<dbl>
1	Control	Fruit	13.5
2	Control	Inv	25.3
3	Control	Plant0	26.4
4	Control	Scav	0.382
5	Control	Seed	34.4
6	Long-term Krat Exclosure	Fruit	3.72
7	Long-term Krat Exclosure	Inv	21.8
8	Long-term Krat Exclosure	Plant0	20.4
9	Long-term Krat Exclosure	Scav	1.15
10	Long-term Krat Exclosure	Seed	52.9

```
[1] "7d"
```

```
# A tibble: 5 x 4
  diet    control exclusion difference
  <chr>    <dbl>    <dbl>    <dbl>
1 Seed    34.4      52.9      18.5
2 Scav     0.382    1.15     0.771
3 Inv     25.3      21.8     -3.49
4 PlantO  26.4      20.4     -6.02
5 Fruit   13.5       3.72    -9.77
```

```
[1] "7e"
```



8: Do the Database Masses Match What Portal Measured?

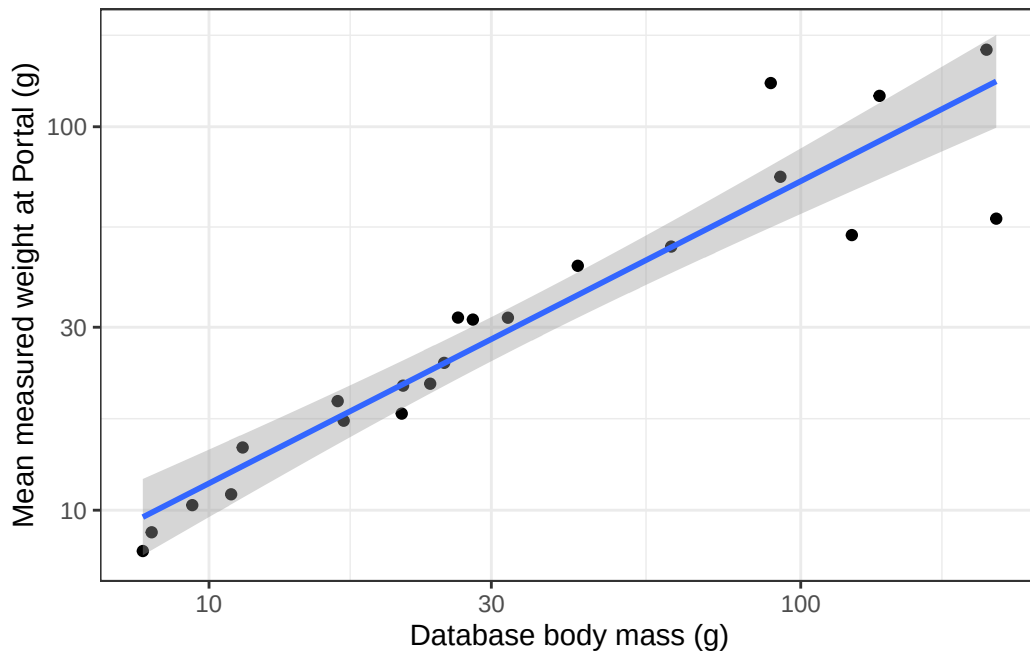
```
[1] "8a-b"
```

```
# A tibble: 22 x 4
  sci_name          mass_g n_captures mean_weight_g
  <chr>            <dbl>    <int>    <dbl>
1 Baiomys taylori      8        38      8.76
2 Chaetodipus baileyi 26.4     1836     31.8
3 Chaetodipus hispidus 32        22     31.8
```

4	<i>Chaetodipus intermedius</i>	16.5	9	19.2
5	<i>Chaetodipus penicillatus</i>	16.9	1987	17.1
6	<i>Dipodomys merriami</i>	42	7005	43.4
7	<i>Dipodomys ordii</i>	60.4	1756	48.7
8	<i>Dipodomys spectabilis</i>	136.	1415	120.
9	<i>Neotoma albigula</i>	206.	776	159.
10	<i>Onychomys leucogaster</i>	27.9	579	31.4

i 12 more rows

[1] "8c"



[1] "8d"

```
# A tibble: 22 x 5
  sci_name      n_captures mass_g mean_weight_g ratio
  <chr>          <int>   <dbl>         <dbl> <dbl>
1 Sigmodon fulviventer      38    214          57.6  3.72
2 Sigmodon ochrognathus     31    122          52.2  2.34
3 Neotoma albigula       776   206.          159.  1.30
4 Sigmodon hispidus      119    92.4          74.0  1.25
5 Dipodomys ordii      1756    60.4          48.7  1.24
6 Peromyscus leucopus        7    21.2          17.9  1.19
```

7	<i>Dipodomys spectabilis</i>	1415	136.	120.	1.13
8	<i>Peromyscus eremicus</i>	557	23.6	21.4	1.11
9	<i>Onychomys torridus</i>	1381	25.0	24.2	1.03
10	<i>Chaetodipus hispidus</i>	22	32	31.8	1.01

i 12 more rows